Assessment 6:Time Series Analysis

Perform the following steps using the built-in AirPassengers dataset in R:

1 .Load the built-in AirPassengers dataset in R.

2. Explore the first five and last five observations of the dataset.

3. Identify the starting and ending time periods of the dataset. (start, end)

4. Determine the frequency and time-series characteristics of the dataset. (frequency, class)

5. Visualize the monthly passenger trend over the entire period. (plot)

6. Decompose the time series into trend, seasonal, and random components. (decompose)

7. Plot the decomposed components to examine the underlying patterns.

8. Analyze the autocorrelation and partial autocorrelation of the time series. (acf, pacf)

9. Fit an ARIMA model to the AirPassengers time series. (arima)

10. Display and interpret the fitted ARIMA model parameters and model summary. (summary)

11. Forecast the number of passengers for the next 12 months using the fitted ARIMA model. (predict)

12. Visualize the actual and forecasted passenger values with appropriate labels and title. (plot, lines)